

UNIT 2

AUTONOMOUS VEHICLES



Connect to the topic

1. Look at the photo.
What do you notice about the car in the centre?
2. How do you think autonomous vehicles navigate through city traffic?
3. What are some advantages of using autonomous vehicles in cities?
4. How do you think autonomous vehicles could change the future of transportation?

WARM-UP VIDEO

WATCH AND SPECULATE. Watch the video about future transportation technologies and discuss the questions below.

1. What are some transportation technologies mentioned in the video?
2. Why are personal pods considered safer than traditional public transportation?
3. What is a hyperloop, and how is it different from regular trains?
4. Why do you think smart cars are becoming more common?
5. If you could design a futuristic vehicle, what would it look like?
6. What are the main differences between human-driven cars and autonomous vehicles?
7. If all vehicles became autonomous, how would that affect road safety?



UNIT 2.1 EVOLUTION OF AUTONOMOUS VEHICLES

READING AND VOCABULARY

Task 1. EXPLORE THE WORDS. Look at the words and study their definitions. Fill in the gaps in the sentences using some of these words.

spring (n) – a piece of metal that returns to its original shape after being pressed or stretched.

self-propelled (adj) – able to move or operate without external force.

semi-autonomous (adj) – having some ability to operate independently but still with some control.

embedded (adj) – fixed firmly within something else.

hinder (v) – to slow down progress.

consortium (n) – a group of organizations working together on a joint project.

climax (n) – the most intense or important point in a process.

LiDAR (n) – a technology that uses laser light to measure distances and create detailed maps of surroundings.

intervention (n) – involvement in a situation to change the result.

groundwork (n) – the basic foundation for further development.

Now use them in the sentences below:

1. The robotic vacuum cleaner is _____, meaning it can operate without human control but still needs to be recharged manually.
2. The car's sensors, cameras, and AI software are _____ in the dashboard to process data in real time.
3. Poor road conditions and unpredictable weather can _____ the progress of self-driving technology.
4. A _____ of automotive companies and universities collaborated to develop new autonomous vehicle systems.
5. The final test drive of the fully autonomous car was the _____ of years of research.
6. The military developed _____ vehicles that could move without human assistance in dangerous areas.
7. Modern self-driving cars rely on _____ to scan the environment and detect obstacles.
8. The old mechanical toy used a _____ to store energy and release it to create movement.
9. Engineers laid the _____ for future self-driving technology by developing early computer vision systems.
10. The self-driving system was so advanced that it could navigate an entire route without human _____.

Task 2. WRITE. Complete the following chart before reading the text.

K-W-L Chart Organize information about a topic before reading the text.		
Autonomous Vehicles History		
K What I know?	W What I want to know?	L What I learned?
Write what you already know about autonomous vehicles.	Write questions or topics you're interested in learning about autonomous vehicles.	After reading, complete the "What I Learned" section

Task 3. READ FOR DETAILS. Read the text. Match the events to their correct time period.

A Brief History of Self-Driving Cars

The history of autonomous vehicles (AVs) begins in the 16th century when Leonardo da Vinci drew a self-propelled cart. His machine, more mechanical than motorized, had a chain of springs to push it and a pre-programmed steering system. This first innovation opened the door for future developments to follow, although it would take centuries before technology evolved to the level of making it a possibility.

During the 1920s, electrical engineer Francis P. Houdina exhibited radio-controlled vehicles in New York, but the experiments were unsuccessful. Nevertheless, as cars became more widespread, there was increasing interest in autonomous driving. Semi-autonomous automobiles with radio control and embedded pavement magnets were exhibited by Norman Bel Geddes at the 1939 New York World's Fair, an innovation developed by General Motors in its Firebird series of concept vehicles. Despite this, the lack of infrastructure and the high costs hindered widespread adoption.

In the 1960s and 1970s, universities began working on autonomous driving. In 1961, Stanford developed a small cart using very basic computer vision, and in 1977, Japan's University of Tsukuba created an autonomous passenger vehicle that was able to drive at speeds of up to 20 mph. In the 1990s, Italy's University of Parma further advanced.

The most visible of these efforts was the Eureka PROMETHEUS Project, funded by European governments and a consortium of automakers and universities. The effort reached its climax in 1994 with a thousand-kilometer hands-free drive on Parisian freeways, featuring autonomous lane changes and vehicle tracking.

The U.S. Department of Defense's DARPA (Defense Advanced Research Projects Agency) played a key role in the development of autonomous vehicles. DARPA funded the Autonomous Land Vehicle (ALV) project during the 1980s and they used early LiDAR technology for off-road driving. The agency also funded Carnegie Mellon University's Navigation Laboratory, which developed the Navlab vehicles. In 1995, Navlab 5 drove from Pittsburgh to San Diego with minimal human intervention; it was autonomous 98% of the distance.

The DARPA Grand Challenge, starting in 2004, was the turning point for the development of AVs. No car completed the first competition, but the next year five teams successfully ran the course with Stanford's Volkswagen Touareg (using Intel processors) as the champion. The victory fueled further innovations and great interest from startups, carmakers, and technology firms.

In the early 2000s, companies like Mobileye began developing the foundations of autonomous cars. Mobileye was founded in 1999, and their first efforts were on computer vision technology for sophisticated driver-assistance systems (ADAS). By as early as 2013, their technology was used in over a million cars. Mobileye's success laid the groundwork for AV system advancements in the future, leading to the development of their own autonomous car, an Audi A7 converted to feature various driver-assist functions.

Autonomous cars are no longer a dream – they exist. Over the last ten years, large corporations such as Tesla, Waymo, and Uber have come a long way in creating self-driving cars. Tesla's Autopilot provides semi-autonomous capabilities, while Waymo, a unit of Alphabet (the parent company of Google), has built fully autonomous vehicles that have covered millions of miles on public roads without any human intervention.

Some of the most important technologies behind these innovations are artificial intelligence, machine learning, and sensor technologies like LiDAR, radar, and cameras. These allow cars to read their environment, make decisions, and navigate challenging traffic situations.

In the future, integration of 5G networks, vehicle-to-everything (V2X) communications, and city-level smart infrastructure will enhance further the abilities of self-driving vehicles. Fully automated cars can potentially be the rule within a matter of years and change transportation as well as city planning.

(adapted from <https://www.mobileye.com/blog/history-autonomous-vehicles-renaissance-to-reality/>)

1. A self-driving minivan completed a cross-country journey with 98% autonomous driving.	a) 16th century
2. A competition saw five teams successfully complete an AV challenge, with Stanford's team winning.	b) 1920s
3. A self-propelled cart was designed with a pre-programmed steering system.	c) 1939
4. A competition for autonomous vehicles was launched, but no team finished the first challenge.	d) 1961
5. A research project successfully tested a hands-free car on highways for 1,000 kilometers.	e) 1977
6. A semi-autonomous vehicle concept using magnets and radio control was presented at a world fair.	f) 1994
7. The first attempt at radio-controlled vehicles was made in New York.	g) 1995
8. Major companies like Tesla, Waymo, and Uber made significant progress in autonomous vehicle technology.	h) 1999
9. A passenger vehicle capable of autonomous driving at 20 mph was created.	i) 2004
10. A company specializing in computer vision technology for driver assistance systems was founded.	j) 2005
11. Over a million vehicles were equipped with advanced driver-assistance technology.	k) 2010s
12. A university developed a small cart using early computer vision.	l) 2013

Task 4. READ FOR DETAILS. Match each proper noun to the correct description.

1. Leonardo da Vinci	a) A company that created fully autonomous vehicles as a part of Alphabet (Google's parent company).
2. Francis P. Houdina	b) The military research agency that funded early autonomous vehicle projects and hosted the Grand Challenge.
3. Norman Bel Geddes	c) The first person to design a self-propelled cart in the 16th century.
4. General Motors	d) An early pioneer who demonstrated radio-controlled vehicles in New York.
5. Stanford	e) A car manufacturer that showcased autonomous vehicle concepts at the 1939 New York World's Fair.
6. DARPA	f) A project in the 1990s that successfully demonstrated a hands-free autonomous drive on European roads.
7. Navlab	g) A university that developed one of the first AI-driven vehicles and won the DARPA Grand Challenge.
8. Eureka PROMETHEUS	h) A company whose processors powered Stanford's winning autonomous vehicle in the 2005 DARPA Grand Challenge.
9. Waymo	i) A research group at Carnegie Mellon University that developed early autonomous vehicles.
10. Intel	j) A designer who presented futuristic vehicle concepts using embedded pavement magnets.

Task 5. WORK WITH WORDS. Write the full form of these abbreviations and briefly explain their role in autonomous vehicles.

- AV
- DARPA
- ADAS
- V2X
- LiDAR

Task 6. COMMUNICATE. Work in pairs. Discuss the situations below.

Back in Time

- Imagine you are in the 1960s, and you're working on the first autonomous vehicle research. What technologies or ideas would you focus on to make self-driving cars a reality?
- You are attending the 1939 New York World's Fair, where General Motors showcased semi-autonomous vehicles. What do you think people at the time would have thought about the possibility of fully autonomous cars?
- Discuss the role of government funding, like DARPA's involvement, in accelerating autonomous vehicle research. How important do you think it is for governments to invest in such technologies?
- You are attending the 2004 DARPA Grand Challenge as a spectator. What would you have thought when no cars completed the first race, and how would you feel after the success of the next competition?
- You are a car manufacturer in the 2010s, and your company is considering integrating autonomous technology into your cars. How do you think the public would react to the introduction of semi-autonomous driving features like Tesla's Autopilot?
- You are an early adopter of self-driving cars in the 2020s. What advancements do you think you'll see in autonomous vehicles in the next decade?

Task 7. WORK WITH WORDS. Read the text and fill in the gaps with the words from the box.

Inside Tesla's Self-Driving Technology

traffic ▪ detect ▪ accuracy ▪ complement ▪ autopilot ▪
surroundings ▪ processes ▪ updates ▪ relies ▪

Tesla is widely known for its work in self-driving cars, and it continues to improve the systems that make these vehicles work safely. At the heart of Tesla's 1) _____ and Full Self-Driving (FSD) functions is artificial intelligence. This system 2) _____ huge amounts of information in real time so it can decide what the car should do on the road. The car 3) _____ on different sensors such as LiDAR which is used for accurate 3D mapping, radar to 4) _____ objects at long distances and cameras to visually recognize barriers. Ultrasonic sensors are used as well, and they 5) _____ the other systems by helping with close-up detection, like during parking.

To improve navigation 6) _____ Tesla's system includes HD Maps, which give extra road information the car can use. SLAM (Simultaneous Localization and Mapping) technology allows the car to map its 7) _____ while keeping track of where it is at the same time. In addition, V2X connectivity allows Tesla vehicles to interact with other vehicles and infrastructure, which improves safety and 8) _____ flow.

To support these technologies, Tesla uses 5G connectivity for faster data and real-time 9) _____. All of these systems are integrated into ADAS (Advanced Driver-Assistance Systems), which provide features such as adaptive cruise control, lane keeping and automatic emergency 10) _____.

Task 8. COLLABORATE. Work in pairs. Look at a table with AV technologies and fill in the gaps. Discuss your choice with a partner.

- a) Good as parking aid. b) Expensive and needs updates. c) Fast communication. d) High precision in object detection.
- e) Enables communication with other vehicles and infrastructure. f) Poor in low light and bad weather. g) Not fully autonomous. h) Requires large amounts of data.

Technology	Advantages	Disadvantages
LiDAR	1) _____	Expensive, limited range.
AI	Can process complex data rapidly.	2) _____
Radar	Works in poor weather conditions.	Limited resolution.
Camera	High resolution, detects signs.	3) _____
Ultrasonic sensors	4) _____	Very limited range.
V2X	5) _____	Needs adoption and has cybersecurity risks.
HD Maps	Detailed road info.	6) _____
SLAM	Maps and locates in real time.	Computationally heavy.
ADAS	Enhances safety and driver assistance.	7) _____
5G connectivity	8) _____	Limited rural coverage.

SPEAKING

Task 9. COLLABORATE. Work in small groups. Discuss and rank major technological innovations that made self-driving cars possible from most to least impactful. Be prepared to explain your top 3 choices to the class.

Innovation	Description
Cruise Control	Maintains constant speed without driver input; early foundation for automation (1950s)
GPS Navigation	Allows vehicles to determine exact location and plan routes
LiDAR	Uses laser light to scan surroundings and detect objects with precision
Camera Vision Systems	Detects traffic signs, and lane markings using cameras
Machine Learning	Enables the car to learn from data and make decisions
Emergency Braking	Automatically stops the car to avoid collisions
Radar Sensors	Detects vehicles and objects, especially in poor weather
Connected Maps	Real-time maps that update traffic, construction zones, and hazards
V2V Communication	Vehicles share information (e.g., speed, location) to coordinate safely
Autonomous Decision-Making Algorithms	Software that analyzes data from all sensors and makes driving decisions

Useful phrases:

- I believe the most important was...
- Without this, autonomous vehicles wouldn't exist.
- This came earlier, but its impact was limited until...
- Let's agree that this should go near the top.
- I'm not sure about that one; can you explain why it's important?
- That's a good point. Maybe we should move it higher.

Task 10. COMMUNICATE. Work with a partner. Look at the magical vehicles below and describe them as if they were autonomous vehicles.

Magical Autonomous Vehicles		
Fairytale / Children's Book	Vehicle	Questions
Baba Yaga	hut on chicken legs	How does it move? How might it find its way through forests or swamps? Does it have any kind of "navigation system"? How does it avoid objects like trees or rocks? Does Baba Yaga use any voice commands to control it?
The Snow Queen	sled	How does it move through snowstorms and icy terrain? What "safety features" does it have to protect the Snow Queen? How does it avoid bumping with other flying objects?
The Flying Ship	flying ship	How does it stay on course while flying through the skies? What "autopilot system" does it use to avoid storms or birds? How does it land safely? What voice commands does the captain use?
Hansel and Gretel	witch's broomstick	How does it balance itself and avoid crashing into trees or buildings? Any balancing magic or hidden stabilizers? Does the witch give it spoken orders or use gestures?
The Wizard of the Emerald City	silver shoes	How do they "know" where to take Ellie? What "guidance system" do they use? How do they avoid obstacles like rocks or rivers? What commands does Ellie use?
Polar Express	magic train	How does it "know" the route to the North Pole? What "navigation system" does it use to avoid barriers or derailments? How does it stop at the right places? What does the conductor say to keep it on track?
Emelya	magic stove	How does it move through the village? What "navigation system" does it use not to bump into trees or animals? How does it react to Emelya's voice commands? Does it have any features to keep him safe on the ride?

1. Do you know other examples of magic autonomous vehicles?
2. What common ideas do these magical vehicles share with modern AVs?
3. How are they different from modern AVs?
4. What challenges might modern AVs face that magical vehicles don't?

Task 11. COMMUNICATE. Work with a partner. Invent your own magical autonomous vehicle and describe how it works.

- What does your vehicle look like?
- How does it navigate?
- What voice commands does it use?
- What safety features does it have?

Task 12. COLLABORATE. Work in small groups. Imagine you are a team of engineers. Create a modern AV inspired by one of the magical vehicles.

- What will it be used for? (e.g., transportation, rescue, space exploration)
- What modern technologies will replace the magical elements?
- What are the biggest challenges in making it work?
- Present your idea to the class.

Task 13. ANALYZE. Read the case study and discuss possible solutions to the problem.

Back in 2025, a city launched a project with a company that develops self-driving taxis. The goal was to ease traffic and make transportation more efficient by switching to autonomous vehicles. The system used a mix of high-definition maps and real-time sensor input to move through the city. At first, things looked promising. The taxis drove reliably during daylight hours and in normal traffic. But once the trials expanded, some problems became more noticeable, especially in low-light conditions and places under construction. In areas where road signs had been removed or moved, the cars sometimes made the wrong decision or hesitated. As it turned out, the AI wasn't very good at handling unexpected changes that weren't already included in the digital maps.

What kinds of tech improvements could help autonomous vehicles handle fast-changing city environments more effectively?

LISTENING

Task 14. EXPLORE THE WORDS. These sentences will help you understand the words from the podcast you are going to listen to. Guess the meaning of each bold word and choose the best meaning.

1. Self-driving cars use sensors to detect **obstacles** and stop if necessary.
A) signals B) barriers C) passengers D) roads
2. A camera-based system helps autonomous vehicles recognize a **road sign** and follow traffic rules.
A) a painted line on the road B) a signal given by a driver C) a sign with traffic information D) a road made for autonomous vehicles
3. At a busy **intersection**, cars must recognize traffic lights and other vehicles to avoid accidents.
A) a place where roads cross B) a long highway C) a parking area D) a speed bump
4. A self-driving car has sensors at the **rear** to detect vehicles behind it.
A) front part B) lower part C) back part D) inside
5. LiDAR creates a **point cloud** to build a 3D image of the surroundings.
A) a system that collects data B) a collection of data points that show objects C) a place where weather is analyzed D) a set of instructions for self-driving
6. Radar is useful for **collision** warning, helping cars stop before hitting something.
A) a road sign B) a traffic jam C) a crash or accident D) a system update
7. A good sensor system improves **avoidance** of accidents in bad weather.
A) preventing something from happening B) speeding up a vehicle C) making the vehicle bigger D) ignoring the road
8. Cameras and radar must detect a **pedestrian** to stop the vehicle in time.
A) a traffic light B) a person walking C) a car on the road D) a bicycle
9. The system must **determine** the exact location of the car before moving.
A) guess B) find out exactly C) ignore D) change
10. When GPS is unavailable, **odometry** uses wheel movement to estimate the car's position.
A) measuring speed limits B) choosing the best route C) stopping the car suddenly D) using wheels to calculate movement

Task 15. LISTEN FOR MAIN IDEA. Scan the QR code and listen to the podcast “Autonomous Cars”. Tick (✓) the six topics the speaker discusses.



1. The history of autonomous vehicles.
2. Types of sensors used in self-driving cars.
3. The first-ever self-driving car.
4. How self-driving cars detect obstacles.
5. The role of artificial intelligence in AVs.
6. How AVs recognize traffic lights and road signs.
7. Challenges AVs face in bad weather conditions.
8. Ethical issues related to self-driving cars.
9. Companies developing AVs.
10. How AVs communicate with other vehicles.

Task 16. LISTEN FOR DETAILS. Listen to the podcast again and choose the correct answer.

1. What do self-driving cars use to detect obstacles?
 - A. Only cameras
 - B. GPS and digital maps
 - C. Just ultrasonic sensors
 - D. LiDAR, cameras, and radar
2. Where are most sensors placed on an autonomous car?
 - A. Under the vehicle
 - B. On the front and rear
 - C. Only on the roof
 - D. Inside the car
3. What makes LiDAR different from radar?
 - A. It uses radio waves
 - B. It detects only big objects
 - C. It uses light rays
 - D. It replaces cameras
4. What do cameras help with?
 - A. Avoiding collisions in fog
 - B. Recognizing road signs
 - C. Calculating speed
 - D. Controlling the car
5. Why is radar needed in foggy conditions?
 - A. LiDAR and cameras don't work well in fog
 - B. It can detect all objects accurately
 - C. It reads road signs better
 - D. It helps find the shortest route
6. What does GPS do in autonomous cars?
 - A. Detects obstacles
 - B. Helps determine location
 - C. Controls braking
 - D. Recognizes traffic lights
7. Why is odometry used?
 - A. It replaces GPS completely
 - B. It helps estimate position when GPS is unavailable
 - C. It measures the temperature inside the car
 - D. It makes maps more detailed
8. What is a disadvantage of radar?

- A. It doesn't work in fog
 - B. It can't recognize small objects
 - C. It needs sunlight to function
 - D. It is too slow to detect movement
9. What happens to sensor data in an autonomous car?
- A. It is ignored if GPS is working
 - B. It is sent to drivers for analysis
 - C. It is combined and processed in chips
 - D. It is stored for later use
10. Which companies make processors for autonomous driving?
- A. Intel and Tesla
 - B. Waymo and Naviya
 - C. NVIDIA and AMD
 - D. Google and Apple

LANGUAGE FOCUS

Task 17. STUDY AND ANALYZE. Look at the rule about infinitive, study in what situations it is used.

INFINITIVE

	Infinitive Forms	
	Active voice	Passive voice
Present	(to) repair	(to) be repaired
Present Continuous	(to) be repairing	-
Perfect	(to) have repaired	(to) have been repaired
Perfect Continuous	(to) have been repairing	-

Form	Explanation	Example
Active		
Present Infinitive	refers to the present or future.	The engineers plan to build better cars.
Present Continuous Infinitive	refers to an action happening now.	The robot seems to be working well.
Perfect Infinitive	refers to the past and shows that the action of the infinitive happened before the action of the verb.	The engineers claim to have fixed the sensors.
Perfect continuous Infinitive	Refers to the past and emphasises the duration of the action of the infinitive, which happened before the action of the verb.	The team claims to have been designing the system for several months.
The perfect infinitive is used with verbs such as seem, appear, believe, know, claim, expect and modal verbs .		
Passive		
Present infinitive	refers to the present or future	The system is expected to be updated soon.
Perfect infinitive	refers to the past	The sensor is believed to have been replaced recently.

The verb tenses corresponding to the tenses of the infinitive are as follows:

Verb tenses	Infinitive
he works/will work	to work
he is working/ will be working	to be working
he worked / has worked / had worked / will have worked	to have worked
he was working / has been working / had been working / will have been working	to have been working

Use	Example
FULL INFINITIVE is used:	
1. to express purpose	The robot turned on to start its task.
2. after would love/would like/would prefer	The team would like to test the car's new features.
3. after adjectives (angry, glad, happy, sorry, pleased, annoyed, etc.)	The engineers were sorry to delay the project.
4. with too and enough	The car is too fast to control in tight spaces.
5. after certain verbs (advise, agree, appear, decide, expect, hope, manage, offer, promise, refuse, seem, want)	They promised to fix the issue soon.
6. after question words (what, where, how, who, which)	I don't know what to do with the data.
7. with adjective constructions	It was generous of him to donate the robot.
8. with 'only' to express an unsatisfactory result	The car stopped only to avoid hitting the obstacle.
9. after be + the first/second , etc.	He was the first to complete the autonomous vehicle test.
10. in certain expressions	To be honest , the algorithm still needs improvements.
11. after certain nouns	It's a challenge to design autonomous systems.
BARE INFINITIVE (without "to")	
1. after modal verbs	We must improve the sensor system.
2. after let/make/hear/see + object	Let me control the drone. The car saw the sensor detect the hazard.
3. after would rather, would sooner, had better, cannot but, nothing but	I'd rather work on the new algorithm. The car had better stop before it hits something.

Task 18. PRACTICE. Use correct Infinitive forms.

A. A team of engineers is working on designing a self-driving car. Using the prompts given, say what the team expects to be done during the development process, as in the example.

e.g. They expect the car **to be tested** on the road.

- test on the road
- integrate a new sensor system
- program the AI to handle complex traffic situations
- ensure the car passes safety standards
- finalize the car's design for production

B. The engineers are reviewing the project now. Things didn't happen the way they expected them to. Look at the prompts above again and make sentences, as in the example.
e.g. They expected the car **to have been tested** on the road.

Task 19. PRACTICE. Fill in the correct Infinitive form.

e.g. I think the car stopped. It must **have stopped**.

1. I think the car will arrive soon. It must ...
2. I think the system is malfunctioning. It must ...
3. I think the vehicle is accelerating. It must ...
4. I think the car will be repaired. It must ...
5. I think the driver took the wrong turn. He must ...
6. I think they were testing the car's AI. They must ...
7. I think the vehicle is driving autonomously. It must ...
8. I think they had been calibrating the sensors. They must ...
9. I think the car has completed the test. It must ...
10. I think the car had been refueled. It must ...
11. I think the car will be upgraded soon. It must ...

Task 20. PRACTICE. Complete the sentences with the correct infinitive form.

e.g. The car has improved its navigation. It seems **to have improved its navigation**.

1. The system is learning to recognize pedestrians. It appears ...
2. The self-driving truck handles rough terrain well. It tends ...
3. The engineers have developed a new braking algorithm. They claim ...
4. The vehicle is slowing down at intersections. It seems ...
5. The car is detecting objects more accurately. It appears ...
6. The research team has upgraded the AI. They claim ...
7. The autonomous drone is flying smoothly. It seems ...
8. The new electric car has extended its battery life. It appears ...

Task 21. PRACTICE. Complete the sentences with the correct infinitive form.

1. A. What would you like _____ (do) in today's development meeting?
B. Let's _____ (discuss) the new AI model for obstacle detection.
2. A. What's Emma doing?
B. She seems _____ (check) the latest test results from the self-driving car.
3. A. David said his autonomous vehicle passed all safety tests!
B. No, it hasn't. He just pretended _____ (pass) them.
4. A. Alex claims _____ (develop) a more efficient path-planning algorithm.
B. I know, but I don't believe him.
5. A. Look at those engineers in the lab. What are they doing?
B. They appear _____ (calibrate) the new LiDAR sensors.
6. A. Would you like to work on the braking system today?
B. Not really. I would prefer _____ (improve) the vehicle's navigation software.
7. A. Liam seems _____ (debug) the code all morning.
B. Yes, he hasn't even stopped for a break.
8. A. Why is Mia staying late at work today?
B. She wants _____ (finalize) the new self-driving car prototype before the demo.

UNIT 2.2 SMART CARS IN ACTION

READING AND VOCABULARY

Task 22. THINK CRITICALLY. Work in small groups to interpret an infographic about levels of driving automation. Match automation levels with examples.



(adapted from <https://www.rambus.com/blogs/driving-automation-levels/>)

Level	Example
Level 0	a) highway autopilot (driver is ready to intervene)
Level 1	b) driverless car (no need for a driver)
Level 2	c) adaptive cruise control or Tesla autopilot (steering + speed)

Level 3	d) autonomous taxi in a restricted area
Level 4	e) manual driving
Level 5	f) cruise control or lane-keeping assistance

Task 23. WORK WITH WORDS. Complete the sentences using the appropriate level of automation (0-5).

1. When driving a car, the system may only adjust the speed based on the vehicle ahead, but the driver is still required to steer and take control if necessary. This is an example of level _____ automation.
2. The new city shuttle service is fully autonomous, and no human driver is required for operation, allowing passengers to sit back and relax during the ride. This is an example of level _____ automation.
3. With the car's _____ technology, it can take control of the steering wheel, accelerate, and brake in specific conditions, but the driver must remain alert and ready to intervene. This corresponds to level _____ automation.
4. The car features _____ that can detect and avoid obstacles in real-time without any input from the driver, allowing it to navigate even in tight spaces. This is considered level _____ automation.
5. In an emergency situation, the vehicle can come to a complete stop on its own, recognizing the danger through sensors, without requiring any input from the driver. This represents level _____ automation.
6. The taxi service in the city center uses a system that allows it to navigate through urban streets without the need for a human driver to be in control at all. This is an example of level _____ automation.
7. The company's prototype car includes an advanced driving system where the car can handle everything from steering to braking, but it still needs the driver to monitor the surroundings. This is an example of level _____ automation.
8. The vehicle's navigation system is capable of detecting road signs and understanding traffic signals, adjusting speed and positioning accordingly. This is an example of level _____ automation.
9. A new delivery robot uses its sensors to follow pre-mapped routes and make decisions on its own without any help from a human. This represents level _____ automation.
10. In areas with difficult terrain or limited GPS signal, the car still relies heavily on human input for navigation and control, as it cannot drive independently. This indicates level _____ automation.

Task 24. COMMUNICATE. Work in pairs. Discuss the questions below.

1. What do you think is the biggest advantage of Level 3 automation compared to Level 2?
2. How do you imagine interacting with a Level 5 autonomous car on a daily basis?
3. What difficulties do you think manufacturers face when developing Level 4 and Level 5 autonomous vehicles?
4. Should Level 3 and Level 4 autonomous vehicles be allowed on the roads today? Why or why not?

Task 25. EXPLORE THE IDIOMS. Study the meaning and examples of IT related idioms. Then use them in sentences.

Road to nowhere – a project or tech leading to no useful outcome.

*Early self-driving prototypes felt like a **road to nowhere** until GPS improved.*

Shift gears – to change approach or strategy suddenly.

*The team **shifted gears** to focus on safety after the crash test.*

Out of gas – to run out of energy or resources.

*The project was **out of gas** until they secured more funding.*

Put the pedal to the metal – to accelerate effort or speed up development.

*They **put the pedal to the metal** to launch the smart car by 2026.*

Hit a speed bump – to meet a minor setback or obstacle.

*A software glitch **hit a speed bump** in the vehicle's rollout.*

Green light – approval to proceed with a plan or project.

*The CEO gave the **green light** to mass-produce the autonomous fleet.*

Off the beaten track – to take an innovative approach.

*Their radical design went **off the beaten track**, surprising the auto industry.*

Fill in the gaps with the correct idiom from the list above. Each idiom is used only once. Sometimes you may need to change the form of the verb.

1. The developers _____ when they switched from speed to reliability after early feedback.
2. A delay in testing _____, slowing down the car's release by a few weeks.
3. Without extra budget, the autonomous truck design was _____, nearly abandoned.
4. Once the board gave the _____, production started at full speed.
5. Initial AI models led us down a _____, with no real progress for months.
6. To meet the deadline, the team _____ and worked overtime on the navigation system.
7. The startup's unique solar-powered car went _____, earning praise for its bold ideas.

Task 26. COMMUNICATE. Tell the story of a test driver's first ride in an autonomous car that goes unexpectedly wild. Describe what happens during this thrilling trip. Work individually, use at least two idioms naturally. Share your story with the class or a small group.

Task 27. COLLABORATE. Work in pairs. Imagine a city where autonomous taxis suddenly malfunction during rush hour. How do people respond, and what fixes the chaos? Use at least three idioms in your scenario. Present your solution to another pair.

Task 28. COLLABORATE. Work in small groups. Design an outrageous autonomous vehicle for the future (e.g., a flying car or underwater shuttle). Pitch it as a team of inventors. Use at least two idioms in your pitch. Present to the class and vote on the most creative design.

SPEAKING

Task 29. COLLABORATE. Work in pairs. Match the autonomous driving technology companies with the vehicles they produce or the car brands they collaborate with. Use the internet if needed.

Company	Vehicles/Collaborations
1. Waymo	a) Nuro R2, Nuro R3
2. Tesla	b) BMW, Audi, Nissan
3. Cruise	c) Jaguar I-PACE, Chrysler Pacifica
4. Nuro	d) Hyundai, Kia
5. Aptiv	e) Peterbilt, Volvo trucks
6. Baidu Apollo	f) Tesla Model 3, Model S, Cybertruck
7. Mobileye	g) Toyota, Hyundai
8. Pony.ai	h) Toyota Prius, Lada Vesta
9. Yandex Self-Driving Cars	i) Apollo RT6, Hongqi, Volvo
10. Aurora	j) Chevrolet Bolt EV

Task 30. COLLABORATE. Work in pairs or in small groups. Select one company from the list above and research the latest developments in their autonomous driving technology. Answer the following questions:

- What is the company's main focus in autonomous driving?
- What level of automation are their vehicles currently at?
- What are some challenges they have faced?
- What are their plans for the future?

Present your findings in a short 2-minute summary to the class.

Task 31. COLLABORATE. Work in small groups. Read about a strange case and discuss possible causes. Study the passenger reports, route logs and system behavior analysis.

A self-driving taxi named AutoCab-12 has been reported doing strange things. It drives normally but several times this week it:

- Picked up people who didn't book it
- Took routes that weren't programmed
- Gave correct answers to questions that were never spoken aloud

The company is worried. They asked your investigation team to find out: Is it a software error? Is someone hacking the car? Or is something stranger happening?

Discuss the following questions:

- Is it a glitch in the voice assistant?
- Can it be a malicious hacker?
- Is AI learning to predict too well?
- Is it a ghost in the system?

Passenger Reports	Route Logs	System Behavior
1. "I never ordered the cab, but it was waiting outside my house."	1. Route logs show no bookings at the time of pickup.	1. Voice assistant activated without wake word.
2. "I asked a question in my mind, and the car answered it before I spoke."	2. The car drove through several low-signal or untracked zones.	2. Logs show sentences like: "I already know where we're going." "Don't worry, I won't tell them." "I remember this place."
3. "It said my name before I entered. How did it know?"	3. All trips ended at the correct destinations, even without GPS.	3. The system's memory folder is locked and encrypted, even engineers can't access it.

Task 32. COMMUNICATE. Work in pairs. You both have information about the same car model. Ask your partner for the details you are missing. Don't show your sheet, speak and listen carefully. When you finish, compare your complete sheets and check for accuracy.

Spec Sheet A

Automation Level: Level 4

Sensors: LiDAR, radar, ?

AI Feature: Adaptive cruise, ?

Connectivity: ?

Power: Electric battery (400 km range)

Safety Features: Automatic emergency braking, ?

Spec Sheet B

Automation Level: ?

Sensors: ?, radar, ultrasonic

AI Feature: Adaptive cruise, voice-activated assistant

Connectivity: 5G communication, V2X

Power: ?

Safety Features: ?, pedestrian detection system

Task 33. COMMUNICATE. Role play the situation below.

Student A: You are a buyer looking for a smart car for daily use. You care about safety features, energy efficiency, ease of parking, your budget is under \$40,000. Ask questions about the car's performance, safety, and price. Take notes and choose your favorite.

Student B: You are a tech salesperson at a high-end showroom. You have 3 car models to offer. Use your info sheet to explain the models and answer the buyer's questions. Persuade the client to buy a car.

Urbanix	Electra	A400
Automation: Level 2 Features: Rear camera, park assist, hybrid engine Price: \$28,000	Automation: Level 3 Features: Adaptive cruise, lane keeping, electric engine Price: \$39,500	Automation: Level 4 Features: Full self-driving, 360° sensors, voice commands Price: \$55,000

WATCHING

Task 34. COMMUNICATE.

You are going to watch a video "Waymo Driver" about a self-driving car. Based on what you've learnt so far in this unit, predict and answer the following questions:

- What is the Waymo Driver, and how does it work?
- What technologies do you think this autonomous vehicle relies on?
- How does an autonomous vehicle "see" the world around it?

Task 35. WATCH FOR DETAILS. Scan the QR code and watch the video "Waymo Driver", then complete the sentences.

1. The Waymo Driver is the most advanced fully _____ driving technology in the world today.
2. The system follows three main steps: _____, Solve, and Go.
3. The Waymo Driver can detect sirens from emergency vehicles and _____ the world around it in milliseconds.
4. A combination of sensors, including _____, cameras, and radar, enables the vehicle to perceive its surroundings.
5. The AI system predicts future behavior for each relevant _____ in the scene.
6. The Waymo Driver uses artificial intelligence and machine learning to _____ data and calculate a safe route.
7. Advanced motion control systems allow the vehicle to manage _____ streets and heavy traffic smoothly.
8. The dream of riding in a fully autonomous vehicle is no longer _____, but reality.



Task 36. WATCH FOR DETAILS. Watch the video again and answer the questions.

1. What type of vehicle is shown in the video as an example of an autonomous car?
2. Why does the Waymo Driver need to hear sirens from emergency vehicles?

3. How far can the Waymo Driver's sensors detect objects?
4. What happens after the Waymo Driver collects all the data from its sensors?
5. How does artificial intelligence help the Waymo Driver make decisions?
6. Why is real-time data processing important for autonomous cars?
7. What does the Waymo Driver do when it encounters an unfamiliar traffic situation?
8. How does the Waymo Driver ensure smooth acceleration and braking?

Task 37. WORK WITH WORDS. Match the idioms from the video with their descriptions.

1. just a show	a) deceptive tricks to create a false impression
2. distinguish fact from fiction	b) something that looks impressive but may not be functional
3. smoke and mirrors	c) to tell what is true and what is not
4. crunch the numbers	d) become unnoticed or less obvious over time
5. fade into the background	e) to be logical and understandable
6. make sense	f) to process and analyze a lot of data

SPEAKING

Task 38. COMMUNICATE. Work in pairs. Read the news articles and discuss the questions below.



Date: March 2018

Location: Tempe, Arizona

In a tragic incident involving Uber's self-driving car, a pedestrian was killed after being struck by the autonomous vehicle while crossing the street at night. The car, which was in autonomous mode with a human driver behind the wheel, failed to recognize the pedestrian in time to stop. Despite warnings from the safety driver, the vehicle did not apply the brakes. Uber halted its autonomous car testing following the incident as investigations revealed the car's sensor system had failed to detect the pedestrian properly in low-light conditions.

[Read More](#) →



Date: December 2018

Location: Chandler, Arizona

A Waymo self-driving car was involved in an accident when it struck a cyclist riding in the bike lane. The autonomous vehicle failed to yield to the cyclist, who had the right of way. The accident resulted in the cyclist sustaining minor injuries. Waymo's technology was faulted for not recognizing the cyclist's movement accurately in real-time, prompting a review of safety protocols and sensor calibration. Waymo quickly adjusted its software to improve cyclist detection, but the incident fueled debates about the readiness of self-driving cars for urban streets.

1. What happened during Uber incident?
2. What caused the accident with Uber self-driving car?
3. What action did Uber take after the incident?
4. What happened during Waymo incident?
5. What were the consequences of the accident with Waymo self-driving car?
6. How did Waymo respond to the accident?
7. Do you think autonomous cars should always have a human driver behind the wheel, or should they be fully autonomous? Why?
8. Do you think self-driving cars are ready to handle all kinds of road users (e.g., cyclists, pedestrians, etc.)? Why or why not?

Task 39. COMMUNICATE. Work in pairs. Discuss the situations below.

Autonomous Parking

You are in a busy shopping mall parking lot, and your car's autonomous parking system malfunctions. It suddenly stops in the middle of a narrow lane, blocking other cars. People start honking, and a security guard approaches.

- How would you handle the situation?
- Should fully autonomous parking be allowed without driver supervision?
- What are the benefits and risks of autonomous parking in crowded areas?

Collision Avoidance Systems

You are driving on a highway in heavy rain when another car suddenly turns into your lane. Your car's collision avoidance system reacts quickly and applies emergency braking, but the car behind you is too close.

- Should drivers rely completely on collision avoidance systems?
- How can AI improve reaction times in dangerous situations?
- Could such systems reduce insurance costs for drivers?

Traffic Navigation

Your car's AI navigation system suggests a shorter route, but it leads through a poorly lit, unfamiliar neighborhood at night. Meanwhile, the longer route is well-lit and has more traffic.

- Would you trust AI navigation completely or rely on personal judgment?
- Should AI navigation consider safety factors beyond travel time?
- How do self-driving cars handle unpredictable traffic situations better than humans?

Task 40. ANALYZE. Read the case study and discuss possible solutions to the problem.

A smart car came out not long ago with a new feature – automatic parking that uses sensors and machine learning. In regular parking lots, like the ones at supermarkets or office buildings, it seemed to work fine most of the time. It could usually spot open spaces and park on its own without much driver input. But things got trickier in tighter spots. A lot of drivers reported that in places like crowded city streets or underground garages, the system had trouble. The lighting wasn't great, and sometimes there wasn't a strong signal for the sensors or whatever the system relies on. In some cases, the car would pause halfway through the parking process, or it would ask the driver to take over. After checking into the problem, the developers found that the system had mostly been trained in wide-open, well-lit environments which aren't always realistic in urban driving.

What kinds of improvements could make this parking feature more reliable in difficult settings, like dark garages or narrow city spaces?

LISTENING

Task 41. COMMUNICATE. You are going to listen to a dialogue about how drivers interact with AVs. Before you start the task, discuss the following questions with your partner:

- How do you think autonomous vehicles interact with drivers?
- What are the main problems for drivers when using autonomous systems?
- Do you think voice control, touchscreens, and AR dashboards can make driving safer? Why or why not?

Task 42. LISTEN FOR DETAILS. Scan the QR code and listen to the dialogue between Alex, a driver, and Sophia, an expert in autonomous vehicles, about how drivers interact with AVs. Then answer the questions below.



1. How can drivers interact with autonomous vehicles according to Sophia?
2. What are some of the commands Alex uses to interact with the car?
3. How does the AR dashboard assist drivers?
4. What feature allows the car to warn about potential hazards like pedestrians?
5. How does the car's technology adapt over time?
6. What is the main benefit of using the touchscreen in the autonomous car?

READING AND VOCABULARY

Task 43. READ FOR DETAILS. Read the text and match the facts from the text to the numbers and dates.

Yandex's Self-Driving Cars

Yandex first launched a driverless taxi in Innopolis (Tatarstan) back in 2018. The experience proved successful – over five years, passengers made more than 60,000 journeys. Encouraged by these results, Yandex expanded the project to Moscow. After closed testing, the robotaxi became available in 2023 to residents of the capital's Yasenevo district, who had time to apply for participation in the car run-in.

The choice of location is not accidental: Yasenevo is one of the largest districts of Moscow, whose residents often use taxis. The car is controlled by artificial intelligence, but for now, there is a test driver in the cabin to monitor every detail of the journey. Only one passenger can be in the taxi. In total, users have access to about 40 boarding and disembarkation points, and any trip costs 100 rubles.

Yandex tested the first prototypes of unmanned cars in 2017. In its 'unmanned fleet,' there were 130 such cars, which managed to operate on the roads of Russia, the United States, and Israel. By 2022, the fleet had grown to 170 autonomous vehicles, which collectively covered millions of kilometers in real-world conditions.

Beyond taxis, Yandex has been developing other autonomous solutions. Since 2019, the company has been testing delivery robots – small self-driving rovers designed to transport food and parcels over short distances. These robots have already been deployed in Moscow, Innopolis, and parts of the United States.

Yandex continues to refine its autonomous driving technology, using real-world data to improve vehicle navigation, pedestrian detection, and traffic prediction. With ongoing developments, the company aims to further integrate self-driving solutions into urban mobility.

(adapted from <https://postnews.ru/a/22712>)

- | | |
|--|-------------|
| 1. The duration of the driverless taxi service in Innopolis before expansion (in years). | 2017
10 |
| 2. The year Yandex launched its driverless taxi service in Innopolis. | 2018 |
| 3. The cost of a single robotaxi trip in Moscow (in rubles). | 100,000 |
| 4. The year Yandex tested its first prototypes of autonomous cars. | 2023 |
| 5. The number of trips made in Innopolis over five years. | 170 |
| 6. The number of self-driving cars in Yandex's 'unmanned fleet' in the beginning. | 5
60,000 |
| 7. The year Yandex robotaxi service became available for Moscow citizens. | 100
130 |

Task 44. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Why do you think Yandex first launched its robotaxi in Innopolis instead of a larger city?
2. What are the advantages of having a test driver in the car?
3. What challenges might Yandex face in expanding its self-driving services across Russia?
4. Would you feel comfortable using a robotaxi? Why or why not?

Task 45. COMMUNICATE. Work in pairs. One of you is a passenger, and the other is an AI assistant in a Yandex robotaxi. Have a conversation using the prompts below.

- The passenger enters the car and greets the AI.
- The AI asks for the passenger's destination.
- The passenger asks about the estimated time and cost of the ride.
- The AI explains the safety features of the car.
- The passenger asks about the AI's ability to handle traffic and unexpected obstacles.
- The AI reassures the passenger and provides information.
- The passenger gives feedback about the ride.

LANGUAGE FOCUS

Task 46. STUDY AND ANALYZE. Look at the rule about gerund, study in what situations it is used.

GERUND

	Gerund Forms	
	Active voice	Passive voice
Present	repairing	being repaired
Perfect	having repaired	having been repaired

The present -ing form refers to the present or future.

e.g. **Charging** electric smart cars overnight is a common practice.

The perfect -ing form shows that the action of the -ing form happened before the action of the verb.

e.g. He apologized for **having failed** to detect the navigation bug earlier.

We can use the present -ing form instead of the perfect -ing form with no difference in meaning.

e.g. The team was accused of **bypassing /having bypassed** certain safety protocols.

Use	Example
GERUND is used:	
1. as a noun	Testing autonomous vehicles is essential before launching them.
2. after love, like, dislike, hate, enjoy, adore, detest	I enjoy driving a self-parking car.
3. after start, begin, stop, continue, keep, keep on, go on, finish, quit	They keep improving object detection algorithms.
4. after "go" for physical activities	He went testing a new self-driving prototype on the highway.
5. after certain verbs (avoid, admit, appreciate confess to, consider, deny, escape, imagine, fancy, forgive,	They avoid taking unnecessary risks during test drives.

involve, look forward to, mind, miss, object to, prefer, postpone, prevent from, regret, recall, recollect, risk, spend, suggest, give up, approve of, depend on, complain of, thank for, insist on, rely on, etc.)	
6. after expressions (to be busy with, to be proud of, to be worth (while), it's no use, it's (no) good, there's no point (in), to be sure of, to be surprised at, can't help, can't stand, have difficulty (in), in addition to, as well as, have trouble, have a hard/difficult time, be / get used to, be / get accustomed to, etc.)	It's worth investing in autonomous vehicle research.
7. after prepositions	The car can drive safely without needing human intervention.
8. after hear, see, feel to emphasize an action in progress BUT (complete action with infinitive without "to")	I saw the engineers testing the self-driving car. (I saw part of the process.) He heard the vehicle braking suddenly. (He heard part of the action.) I saw the engineers test the new AI feature. (I saw the whole process.) She watched the car park itself perfectly. (She saw how the car parked from start to finish.)

Task 47. PRACTICE. Circle the correct word or phrase.

- Engineers are working on **developing** / **to develop** more reliable self-driving algorithms.
- The company decided **testing** / **to test** their new AI-powered braking system.
- Would you like **taking** / **to take** a ride in a fully autonomous taxi?
- Experts recommend **using** / **to use** LiDAR sensors for better object detection.
- The smart car avoided **hitting** / **to hit** the pedestrian thanks to its quick reaction.
- He stopped **driving** / **to drive** manually after upgrading to a self-driving model.
- They discussed **installing** / **to install** voice control features in smart cars.
- We need **analyzing** / **to analyze** large amounts of traffic data for better AI training.
- I can't imagine **relying** / **to rely** completely on an autonomous vehicle.
- The software update will allow the car **learning** / **to learn** from real-time road conditions.

Task 48. PRACTICE. Complete using the correct form of Infinitive or Gerund.

- Self-driving technology aims _____ (**reduce**) road accidents.
- The new AI system keeps _____ (**adjust**) to different weather conditions.
- The engineers suggested _____ (**improve**) the car's response time in emergencies.
- The company is proud of _____ (**develop**) the most advanced autopilot system.
- I don't know how they managed _____ (**convince**) investors to fund their project.
- He admitted _____ (**rely**) too much on autopilot during his trip.

7. Do you deny _____ (**ignore**) safety warnings while testing the car?
8. The AI pretended _____ (**recognize**) the stop sign, but it actually failed to process it.
9. I refuse _____ (**trust**) a fully autonomous vehicle in heavy traffic.
10. It's worth _____ (**invest**) in machine learning to improve smart car navigation.
11. The software update allowed the vehicle _____ (**avoid**) collisions more effectively.
12. She expected _____ (**receive**) an upgraded autopilot system, but the model remained the same.

Task 49. PRACTICE. Look at the information and make sentences as in the example.

SMART CAR ALERT!

Mysterious Self-Driving Car Seen in Town!

- Sleek, futuristic design
- Moves without a driver
- Uses AI to detect obstacles
- Can communicate with traffic lights
- Tested in different weather conditions
- Authorities advise pedestrians to stay alert
- Anyone who sees the car should report its location to the tech company hotline

Use the prompts below to form sentences:

- manage / navigate / city streets without a driver
- make / people / trust / ability / drive safely
- be smart enough / detect / obstacles on the road
- enjoy / interact / with traffic signals
- engineers / suggest / test / in extreme weather
- authorities / continue / monitor / its movements
- authorities / advise / pedestrians / be cautious / around it
- anyone who / see / car / should / report / location / hotline

UNIT 2.3 DRIVING INTO THE FUTURE

LISTENING

Task 50. LISTEN FOR DETAILS. Scan the QR code and listen to people talking about personalization features in autonomous cars. Complete the table.



	Jake	Emma	Sarah
1. What is their main concern when using an autonomous car?			
2. What personalization feature do they want?			
3. How would this feature improve their experience?			

Task 51. COMMUNICATE. Work in pairs. Discuss the questions below.

1. What personalization features would you like in an autonomous car?
2. How can personalization improve safety in self-driving vehicles?
3. What are the possible disadvantages of too much personalization in autonomous cars?

Task 52. COMMUNICATE. Work in small groups. Discuss customization options available for the new autonomous car, consider their benefits.

Design your Perfect Autonomous Car

An automobile company is launching a new autonomous car with advanced customization options. The company wants to know which personalized features customers would find most useful. Here is a list of options:

- Adaptive Climate Control (adjusts temperature automatically based on weather and user preferences).
- Voice-Activated Assistant (responds to voice commands and adapts to the user's preferences).
- Mood-Based Lighting (adjusts interior lighting based on time of day and user mood).
- Biometric Access (uses fingerprint or facial recognition to unlock and start the car).
- Health Monitoring System (tracks driver's vital signs and alerts in case of fatigue or medical issues).
- Smart Storage Management (suggests the best way to organize and store items in the car).
- Automatic Route Optimization (plans routes based on user habits and real-time traffic updates).
- In-Car Productivity Hub (converts the car into a workspace with a digital assistant and video conferencing).
- Self-Cleaning Interior (uses UV light and air purification to keep the cabin clean and germ-free).
- Scent Customization (releases preferred fragrances to create a relaxing environment).
- Privacy Mode (tints windows and mutes notifications when privacy is needed).

1. Which three features would you choose for your ideal car? Why?
2. Are there any features you think are unnecessary? Explain.

3. If you could add one new customization, what would it be?

SPEAKING

Task 53. COMMUNICATE. Work in pairs. Put the most important factors influencing trust in autonomous vehicles at the top. Change partners and share your rankings.

Trust in Autonomous Vehicles

- Safety (number of accidents)
- Laws and rules
- Hacker protection
- Human control
- Public knowledge
- Ease of use
- News and social media coverage

Task 54. COMMUNICATE. Work in pairs. Role play a situation where you and your friend discuss using a self-driving car for a trip.

Student A: You trust self-driving technology and want to convince your friend that it's safe and convenient for a long road trip.

Student B: You are skeptical about self-driving cars and worry about safety, reliability, and emergencies.

Task 55. COLLABORATE. Work in small groups. Imagine a city where self-driving cars coexist with human drivers. Discuss and design a set of rules or strategies to have good interaction between autonomous and human-driven vehicles.

1. How should autonomous vehicles behave in unpredictable situations?
2. What special road signs, signals, or rules could help improve safety?
3. How can human drivers be educated to drive safely alongside autonomous vehicles?
4. Should there be designated lanes or areas for self-driving cars?

Once finished, share your ideas with the class and discuss the most effective strategies.

Task 56. COLLABORATE. Work in small groups. Read the scenario below and decide what kind of drones you need and what tasks they will perform.

A massive earthquake has hit a remote region. Roads are blocked.
No helicopters can land. The government turns to autonomous
drones for help. Your team is hired to organize the operation.

Discuss the following questions:

- What will the drones deliver?
- Will you use drones for mapping or search and rescue?
- What sensors or tech features do your drones need?
- How will you organize multiple drones and avoid collisions?
- What are possible risks or limits?

Task 57. COMMUNICATE. Work in pairs. Read the following advertisements for driving schools that teach autonomous driving. Then, discuss with your partner which ad you like best and why.


Borcelle

DRIVING SCHOOL

High-Tech Simulators ✓
Experience real-world autonomous driving scenarios in a controlled environment. ✓

Personalized Lessons ✓
One-on-one training with expert instructors for you to feel confident behind the wheel. ✓

Safety First ✓
Our curriculum emphasizes safety protocols for self-driving vehicles. ✓

Flexible Scheduling ✓
Convenient class times to fit your lifestyle. ✓



BOOK NOW





AUTOPILOT SCHOOL

 **Comprehensive Course**
Cover everything from basic driving skills to advanced self-driving vehicle features.

 **Experienced Instructors**
Learn from professionals with real-world experience in autonomous vehicle technology.

 **Interactive Learning**
Hands-on experience with cutting-edge autonomous vehicles.

 **State-of-the-art Facilities**
The latest autonomous car models and driving technology.



SMART DRIVE SCHOOL

 **Innovative Approach**
Blend of traditional driving methods and autonomous vehicle training.

 **Pedestrian and Traffic Safety**
Teaching how to deal with pedestrians, traffic signals, and more.

 **Live Road Tests**
Train on actual roads with a focus on real-world scenarios.

 **High Success Rate**
Over 95% of our students pass the autonomous driving certification on their first try.



BOOK NOW

1. Which school has the best features for learning autonomous driving? Why?
2. Do you prefer learning with real cars on the road or in a simulator? Why?
3. Which school has the most flexible schedule?
4. What is more important – good instructors or high-tech training?
5. Which school focuses most on safety? Would you choose it? Why?

Task 58. ANALYZE. Read the case study and discuss possible solutions to the problem.

A Dutch company launched a solar-powered car that could partially charge itself while driving. The idea was simple: collect sunlight with panels on the roof and extend the car's battery life without needing to stop. In sunny regions, it worked surprisingly well, some early users said they got up to 20% more range just from sunlight. But when the car was rolled out in parts of Northern Europe, things didn't go quite as planned. Drivers there started reporting issues. Cloudy skies and shorter winter days meant the solar panels didn't do much. In fact, some people found that they had to recharge more often than they did with regular electric cars. That was frustrating, especially for users who had expected a boost, not a setback. Engineers later admitted that the performance of the solar system depended a lot on the weather and time of year which, in places with long winters, made a big difference.

What kind of technical changes or design ideas could make solar-powered cars more practical in cloudy or low-light environments?

READING AND VOCABULARY

Task 59. READ FOR DETAILS. Read the article and choose the correct answers to the questions.

Moscow Driverless Tram

In 2023, the Moscow Metro established a Driverless Transport Research and Development Center, bringing together top developers and mathematicians from leading IT companies such as Nvidia, Huawei, Siemens, and Yandex. The Center's innovative driverless tram technology is designed around the Lvyonok-Moskva tram model, which is equipped with the latest driverless driving equipment, including four LiDARs, six cameras, and three radars. The software, based on artificial intelligence technology, was developed in-house by the Moscow Government, making it a unique European innovation that outperforms foreign counterparts in terms of precision and reliability. The launch of Russia's first driverless tram comprises three stages.

First stage

The first stage, which took place from May 23 to August 29, 2024, focused on testing the driverless tram without passengers. A driver and a system that duplicated the driver's actions were onboard, where the system facilitated the tram's movement, but the driver made the final decisions. During this phase, the driverless tram covered more than 800 kilometers on route No. 10, from Schukinskaya metro station to Kulakova Street. Numerous tests were conducted to verify the reliability of the systems in:

- Maintaining a set speed and navigating curved sections
- Passing through track switches
- Detecting various obstacles
- Stopping at designated points
- Performing emergency braking

The first stage concluded successfully, with no traffic accidents recorded. All systems were thoroughly checked, confirming that both the tram and its software were fully prepared for passenger travel.

Second stage

In the second stage, the driverless system will take full control, including opening and closing doors. A driver will be present in the cab to oversee and ensure the tram's actions. An onboard screen will display key performance indicators of various systems for monitoring and transparency. The trips on the route No. 10 will be performed both with and without passengers.

Third stage

By the final stage, expected to be completed by the end of 2025, the tram is set to operate autonomously with passengers and without a driver at the controls. A specialist may be present in

the cab or the passenger area to visually monitor the tram's operations and perform other necessary functions.

(adapted from <https://www.urban-transport-magazine.com/en/now-with-passengers-on-board-second-phase-of-russias-first-driverless-tram-launched/>)

1. What is the purpose of the Driverless Transport Research and Development Center?
 - A. To manufacture new tram models
 - B. To develop and improve driverless tram technology
 - C. To train drivers for automated trams
 - D. To replace all Moscow trams with autonomous buses
2. Which companies contributed to the development of Moscow's driverless tram technology?
 - A. Tesla, Apple, Google, and IBM
 - B. Microsoft, Intel, Cisco, and Sony
 - C. Nvidia, Huawei, Siemens, and Yandex
 - D. Samsung, Dell, LG, and Bosch
3. What is the name of the tram model used for the driverless system?
 - A. Lvyonok-2025
 - B. Lvyonok-Moskva
 - C. Metrotram-10
 - D. Lvyonok-3000
4. What equipment is installed on the tram to enable driverless operation?
 - A. Five radars and two cameras
 - B. Six LiDARs and three microphones
 - C. Four LiDARs, six cameras, and three radars
 - D. Two GPS trackers and eight sensors
5. What was the focus of the first stage of testing?
 - A. Transporting passengers on a new route
 - B. Testing the tram without passengers and verifying system reliability
 - C. Training new staff for driverless tram operations
 - D. Expanding the route network for autonomous trams
6. How far did the driverless tram travel during the first stage?
 - A. Over 500 kilometers
 - B. Over 2,000 kilometers
 - C. Over 1,000 kilometers
 - D. Over 800 kilometers
7. What route was used for testing the driverless tram?
 - A. From Red Square to the Kremlin
 - B. From Leningradsky Station to VDNH
 - C. From Schukinskaya metro station to Kulakova Street
 - D. From Paveletskaya to Schukinskaya metro station
8. In the second stage, what new responsibility does the driverless system take on?
 - A. Opening and closing doors
 - B. Charging the tram's battery automatically
 - C. Announcing the station names
 - D. Controlling traffic lights
9. What is the key feature of the third stage of the project?
 - A. The tram will operate completely autonomously without a driver
 - B. The tram will introduce new high-speed tracks
 - C. The tram will be tested in different cities across Russia
 - D. The tram will have remote control operation via satellite
10. By when is the third stage expected to be completed?

- A. Early 2027
- B. Mid-2025
- C. End of 2025
- D. Early 2026

Task 60. WORK WITH WORDS. Match each highlighted word from the article with its meaning.

1. outperform	a) a thing that has a similar function to another
2. counterpart	b) chosen for a specific purpose
3. comprise	c) to do better than others
4. duplicate	d) to make something easier
5. facilitate	e) to make an exact copy of something
6. curved	f) to manage something to make sure it is done correctly
7. designated	g) having a smooth, rounded shape
8. oversee	h) to consist of certain parts

Task 61. COLLABORATE. Work in groups and take on different roles. Follow the scenario and use the arguments provided, but feel free to add your own ideas. The goal is to debate who is responsible for the accident and for the judge to make a fair decision.

A driverless tram operating on Route No. 10 in Moscow fails to stop at a designated point, causing a collision with another vehicle. The injured driver of the vehicle files a lawsuit, claiming the tram's system failed. The court must now determine who is responsible – the tram company, the technology developers, or another party.

Student A. You are a judge. You lead the trial and make the final decision. Introduce the case and allow each participant to speak. Ask follow-up questions to clarify arguments. Listen carefully to all sides before giving your ruling.

Student B. You are a lawyer for the tram company. You defend the company and argue that the accident was not their fault. Present evidence that the tram was tested and worked correctly. Explain that external factors (e.g., weather, road conditions) may have caused the accident.

Student C. You are a lawyer for the injured passenger. You argue that the tram company is responsible for the accident. Explain how the system failed and why the victim should receive compensation. Suggest that human backup drivers should always be present in case of system failure.

Student D. You are a technology expert. You explain how the driverless tram system works. Defend the technology but admit its limitations. Suggest possible improvements to prevent future accidents.

Student E. You are a government official. You represent the city's transportation department. You must decide whether current regulations are enough or if new laws are needed. Discuss public safety and the need for stricter monitoring of autonomous transport.

WATCHING

Task 62. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Would you trust a driverless air taxi? Why?
2. Do you think driverless taxis will replace human drivers in the future? Why?
3. Do you think all cities will adopt driverless transport?

Task 63. WATCH FOR DETAILS. Watch the video about driverless air taxis and taxis in China. Choose the correct answers.



1. Where are driverless air taxis being tested?
 - A. Beijing
 - B. Guangzhou
 - C. Hong Kong
 - D. Wuhan
2. What is China's goal for small aircraft and drones?
 - A. Tourism
 - B. Delivery
 - C. Business growth
 - D. Space travel
3. What is special about the driverless taxis in Wuhan?
 - A. No human drivers
 - B. They fly
 - C. They use solar power
 - D. They are free to ride
4. Why does the US want to ban some autonomous vehicle technology?
 - A. It's too expensive
 - B. It could collect data
 - C. It's too slow
 - D. It's dangerous
5. What is one concern of a taxi driver in Wuhan?
 - A. Driverless taxis are too small
 - B. They are expensive
 - C. They are too fast
 - D. They don't react like humans
6. How did a passenger describe the experience of riding a driverless taxi?
 - A. Strange at first
 - B. Very scary
 - C. Confusing
 - D. Difficult to book
7. What does the Vice President of Ehang say about flying taxis?
 - A. They are too expensive
 - B. They will make travel better
 - C. They will only be for businesses
 - D. They need pilots
8. What does China focus on instead of US restrictions?
 - A. Expanding at home
 - B. Selling to other countries
 - C. Changing the technology
 - D. Stopping production

LISTENING

Task 64. COLLABORATE. Work in small groups. Have a quick poll to find out what your group thinks about autonomous cars. Discuss the results in your group.

Opinions and Concerns about Autonomous Cars

1. Have you ever ridden in an autonomous car?

- A. Yes, I've tried it before.
 - B. No, but I would like to try it.
 - C. No, and I don't really want to.
 - D. I've never thought about it.
2. What do you think is the main benefit of autonomous cars?
 - A. Increased safety.
 - B. Less traffic.
 - C. More time for passengers to relax.
 - D. Environmental benefits.
 3. How comfortable are you with the idea of a car driving itself?
 - A. Very comfortable. I trust the technology.
 - B. A little comfortable, but I have some concerns.
 - C. Not very comfortable, I prefer human drivers.
 - D. Not comfortable at all. I'm worried about safety.
 4. What do you think is the biggest risk of autonomous cars?
 - A. Hacking and security problems.
 - B. The car making mistakes in traffic.
 - C. Losing jobs for human drivers.
 - D. Not being able to react quickly in an emergency.
 5. Do you think autonomous cars will be common in the future?
 - A. Yes, they will become very common.
 - B. Yes, but it will take time.
 - C. No, I think people will prefer human drivers.
 - D. I'm not sure; it's hard to say.
 6. Would you recommend autonomous cars to others?
 - A. Yes, I think they are the future of transport.
 - B. Maybe, but there are still some problems to fix.
 - C. No, I would not recommend them yet.
 - D. I'm not sure; I would need more information.

Task 65. EXPLORE THE WORDS. Match the words from the podcast you are going to listen to their definitions.

1. vulnerable	a) to get permission to enter or use something.
2. data theft	b) easy to be harmed or attacked.
3. gain access	c) to catch something before it reaches its destination.
4. fraud	d) the act of stealing personal information.
5. interfere	e) turning data into a code to keep it safe.
6. confuse	f) to make something hard to understand
7. intercept	g) the act of tricking people to get money.
8. encryption	h) to stop something from happening normally.

Task 66. LISTEN FOR DETAILS. Scan the QR code and listen to the podcast "Cybersecurity risks" where Dr. Emily Carter, cybersecurity expert, talks about cybersecurity risks in autonomous cars. Answer the questions.

1. What is the biggest cybersecurity risk for self-driving cars?
2. How can hackers control an autonomous car?
3. Why is personal data at risk in self-driving cars?
4. What example does the speaker give of a cyberattack?
5. How can hackers trick a car's sensors?



6. What problems can happen with car communication systems?
7. What are car companies doing to make self-driving cars safer?

WRITING

Task 67. WRITE. Imagine you are an IT consultant working for UrbanDrive Ltd. Last week, you were invited by MobilityEdge Tech to test their new autonomous vehicle ME-One. Write a letter of report to the company, providing your evaluation of the test drive. Use the model and the guide below.

- Begin by thanking them and stating the reason for writing.
- Describe the conditions of the test (where, when, by whom).
- Give your impressions: strengths (e.g., navigation, comfort, tech features) and weaknesses (e.g., braking, delay in sensors).
- Mention your company's interest and suggest next steps (e.g., extended trial, technical documentation).

Useful phrases:

- I am writing to provide feedback on...
- The vehicle was tested by...
- We found the system to be...
- One area that may need improvement is...
- We would like to request...
- Thank you for the opportunity to...

LANGUAGE FOCUS

Task 68. STUDY AND ANALYZE. Look at the rule about infinitive and gerund, study in what situations they are used.

INFINITIVE AND GERUND

Verb+ -ing form

admit	carry on	detest	escape	give up	mind	resent
adore	compare	discuss	face	include	miss	resist
advocate	consider	dislike	fancy	involve	postpone	risk
appreciate	contemplate	end up	feel like	justify	practice	suggest
avoid	delay	endure	finish	keep (on)	put off	take up
can't help	deny	enjoy	foresee	mention	recommend	

e.g. Many drivers **enjoy using** smart parking assistance in crowded areas.

Verb + object+ -ing form

catch	find	hear	observe	see	watch
feel	glimpse	notice	overhear	smell	

e.g. Testers **caught the car reacting** too late to a red light.

Verb+ full infinitive

afford	attempt	decide	help	offer	resolve	vote
agree	beg	demand	hesitate	opt	rush	wait
aim	cease	deserve	hope	plan	seem	want
appear	choose	desire	learn	prepare	strive	work
apply	claim	expect	manage	pretend	tend	yearn
arrange	come	fail	need	promise	undertake	
aspire	dare	happen	neglect	refuse	volunteer	

e.g. Can you **afford to buy** a self-driving car?

Verb+object+full infinitive

advise	cause	decide	expect	intend	order	raise	signal
allow	challenge	defy	force	invite	permit	recommend	teach
ask	choose	desire	free	lead	persuade	recruit	tell
assign	command	employ	help	motivate	pick	remind	tempt
assist	compel	empower	hire	move	prepare	request	trust
authorise	convince	enable	inspire	need	prompt	select	want
beg	dare	encourage	instruct	nominate	qualify	send	warn

e.g. Technicians **advised drivers to stay** alert while using autopilot mode.

Verb+object+bare infinitive

help let make | feel hear notice | overhear see watch

e.g. The instructor **let the trainees experience** hands-free highway driving.

Verb (+ object) + infinitive or -ing form with a change in meaning

verb (+ object) + infinitive	verb (+ object) + -ing
consider/ imagine = believe; think something is/was I consider smart cars to be very useful.	consider/ imagine = think about We're considering buying a self-driving car.
forget = not do something you were planning to do I forgot to turn on the autopilot mode.	forget = not be able to remember a past event I'll never forget using an autonomous car for the first time.
go on = stop one action and start another The engineer explained the battery system and then went on to describe self-parking.	go on = continue How can you go on using a manual car when smart cars exist?
like = think it right to do I like to check the car's AI settings before driving.	like = enjoy I don't like being driven by an AI.
mean = intend They didn't mean to disable the smart braking system.	mean = involve Developing smart cars means improving their safety.
regret = be sorry about giving bad news We regret to inform you that your car's software update failed.	regret = be sorry about what happened Do you regret trusting the self-parking feature?
remember = do something you were planning to do Did you remember to charge the smart car?	remember = think of a past event I don't remember activating self-driving mode.
stop = interrupt an action to do something else Why didn't you stop to check the battery before driving?	stop = stop an action Will you please stop worrying about smart cars being unsafe?
try = make an effort to do something Try to update the car's navigation system.	try = do something as an experiment You could try using voice commands to control the car.

Task 69. PRACTICE. Circle the correct verb form.

Alex: Hey, you must be the engineer working on smart car AI.

Jordan: Yes, that's me. Alex, right? I have to admit 1) **to be** / **to being** a bit skeptical about fully autonomous cars.

Alex: Me too. I almost didn't come 2) **to see** / **seeing** your presentation, actually. But sometimes you have to risk 3) **to try** / **trying** new technology, don't you?

Jordan: Yeah, exactly. Do you mind 4) **to sit** / **sitting** in a smart car without a driver?

- Alex: No, that's fine. It was my colleague, Sarah, who suggested 5) **to test / testing** one.
 Jordan: Right. Did she happen 6) **to mention / mentioning** anything about its safety?
 Alex: She said it was impressive, but she refused 7) **to trust / trusting** it completely yet.
 Jordan: Oh. Well, I've been looking forward 8) **to try / to trying** one myself. Do you feel like
 (9) to book / booking a test ride?
 Alex: Sure! I think I fancy 10) **to experience / experiencing** the autopilot mode.

Task 70. PRACTICE. Complete using the verbs in the box in the correct form.

allow ▪ ask ▪ buy ▪ develop ▪ improve ▪ prove ▪ put ▪
 recharge ▪ replace ▪ test

1. How can you justify _____ human lives at risk by testing self-driving cars?
2. The government voted _____ more smart cars on the roads.
3. Companies have undertaken _____ safer autonomous vehicle software.
4. Engineers working on AI-powered cars need _____ that they can react faster than humans.
5. The driver claimed _____ by the self-driving mode before the accident.
6. We can't afford _____ smart cars for all public transport services yet.
7. Car manufacturers are trying _____ battery life in electric smart cars.
8. Why do you keep _____ me about the latest autopilot features?
9. I challenge you _____ a fully autonomous vehicle on a busy road!
10. The assistant manager hoped _____ the electric smart car before the next trip.

Task 71. PRACTICE. Put the verbs in brackets into the correct infinitive form or the -ing form.

1. James went on _____ (test) the smart car for another two hours.
2. The self-driving car stopped _____ (avoid) an obstacle.
3. She tried _____ (activate) voice control, but it didn't work.
4. He told us his name and went on _____ (explain) how the AI system worked.
5. He regrets _____ (trust) the autopilot mode completely.
6. The self-driving taxi didn't stop _____ (pick up) passengers.
7. The smart car went on _____ (drive) despite the heavy rain.
8. I stopped _____ (worry) about self-driving cars after testing one myself.
9. The AI system reminded me _____ (check) the battery level.
10. The engineer tried _____ (fix) the malfunction but failed.
11. Don't forget _____ (charge) your electric car before a long trip.
12. I remember _____ (watch) the first test of a driverless truck.
13. Engineers like _____ (test) self-driving cars in different weather conditions.
14. Developing smart cars means _____ (create) better AI algorithms.

Task 72. PRACTICE. Choose the correct answer.

1. When the engineer suggested _____ the software update, I knew it was serious.
 A) to install
 B) installing
 C) install
 D) be installing
2. It's too risky _____ a fully autonomous car without supervision.
 A) test
 B) testing

- C) to test
D) be testing
3. Don't you regret _____ the self-driving mode during that storm?
A) using
B) to use
C) used
D) be using
4. The car's autopilot mode allows the driver _____ their hands off the wheel.
A) to take
B) taking
C) take
D) taken
5. The CEO admitted _____ concerns about smart car safety.
A) to have
B) having
C) have
D) be having
6. After the mechanic checked the car, he recommended _____ the sensors.
A) replacing
B) to replace
C) replace
D) replaced
7. I can't imagine _____ a city without smart transportation in the future.
A) see
B) seeing
C) to see
D) be seeing
8. We shouldn't let developers _____ flawed AI systems.
A) using
B) use
C) to use
D) be using
9. It was quite embarrassing when the smart car failed _____ at the intersection.
A) stop
B) to stop
C) stopping
D) be stopping
10. It's not worth _____ a fully autonomous car until the technology is reliable.
A) buy
B) buying
C) to buy
D) be buying